Sales Data Reconciliation & Revenue Validation Report

1. Project Overview

This project focuses on validating revenue accuracy by reconciling expected order values with actual customer payments. The analysis was performed end-to-end, including data cleaning, reconciliation logic, business analysis, and dashboard visualization.

The objective is to identify discrepancies, understand their root causes, and provide actionable insights to improve financial accuracy and operational efficiency.

2. Business Problem

Revenue inconsistencies arise between expected order values and actual payments due to multiple items and transactions within a single order.

Without a structured reconciliation process, this can lead to:

- Revenue misreporting
- Financial losses due to underpayments
- Customer trust issues in case of overpayments
- Inefficient decision-making

The core problem is:

Do actual payments accurately reflect the true value of each order?

3. Validation Logic

- **Expected Revenue** = SUM(price + freight_value)
- **Actual Revenue** = SUM(payment_value)
- Orders are classified as:
    - Match
    - Underpaid
    - Overpaid

4. Data Preparation

- Cleaned and standardized data using Python
- Validated schema and relationships
- Handled missing values appropriately
- Integrated datasets into SQL for analysis

5. Reconciliation Analysis

- Revenue mismatch is minimal (~0.018%)
- Discrepancies concentrated in cancelled and shipped orders
- Specific states contribute majority of mismatch
- Top 10 customers contribute <1% (diversified revenue)

6. Dashboard

Interactive Power BI dashboard with KPIs, mismatch analysis, and business insights.

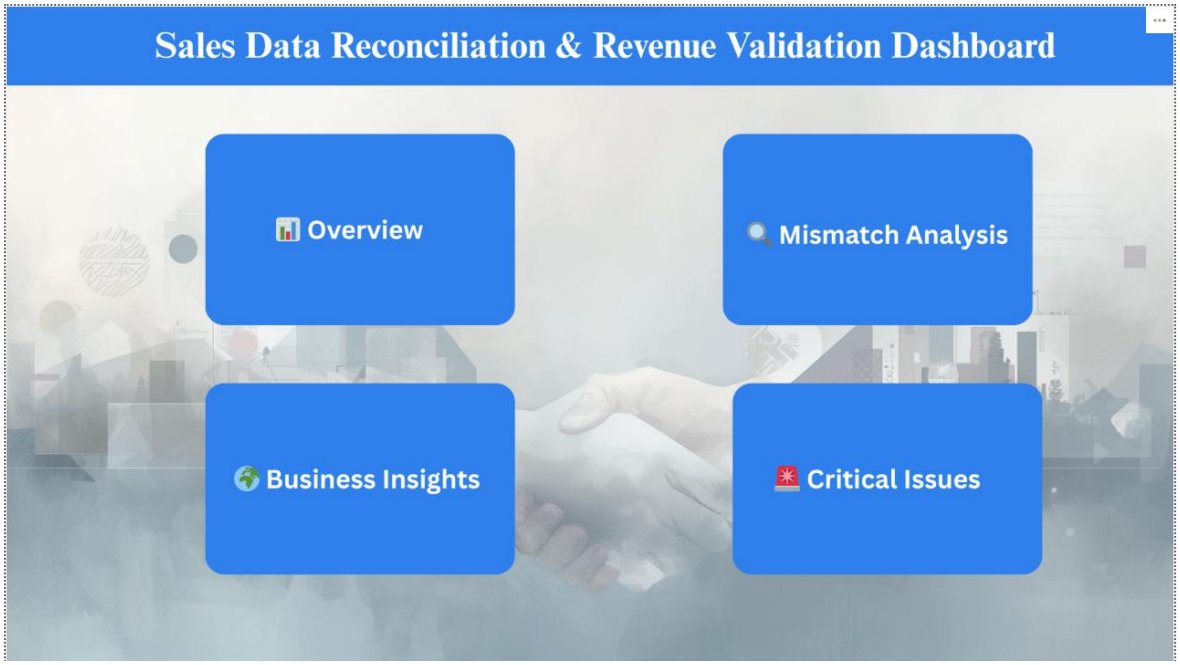

Figure 1: Sales Reconciliation Dashboard Home Page

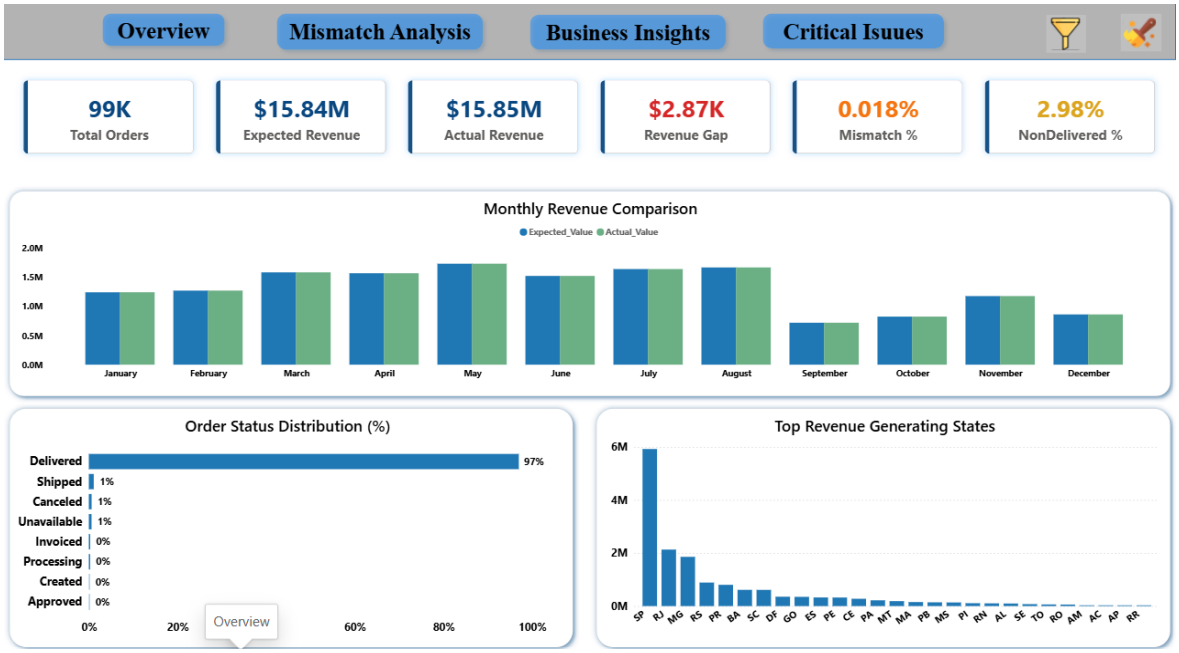

Figure 2: Sales Reconciliation Dashboard Overview

This dashboard provides:

- KPI monitoring (Revenue, Mismatch %, Orders)
- State-wise and order-status mismatch analysis
- Customer contribution insights
- Risk-based conditional formatting

7. Key Insights

- Revenue mismatch is minimal (~0.018%), indicating strong financial alignment
- Discrepancies are concentrated in cancelled and shipped orders
- Specific states contribute the majority of revenue mismatch
- Revenue is highly distributed across customers, with top 10 contributing <1%

8. Business Impact

- Fixing top contributing states can reduce ~60–68% of total mismatch
- Improving reconciliation for key order statuses can significantly reduce discrepancies

- Low customer concentration reduces dependency risk but highlights opportunity to increase customer value

9. Recommendations

- Improve reconciliation process for cancelled and shipped orders
- Focus on high-risk states for operational improvements
- Monitor customers with repeated discrepancies
- Optimize delivery and payment workflows
- Increase average order value through targeted strategies

10. Conclusion

- The analysis confirms that overall revenue alignment is strong. However, localized discrepancies exist across specific order statuses and regions.
- By addressing these high-impact areas, the business can improve financial accuracy, reduce revenue leakage, and enhance operational efficiency.